NYRA Platform - Complete Feature List

🤖 CORE AI AGENT CAPABILITIES

NYRA Super Worker (Flagship Generalist Agent)

- **Research & Analysis**: Comprehensive web research across multiple sources, document analysis, report synthesis, market research, competitive intelligence
- **Browser Automation**: Navigate complex websites, extract data from multiple pages, fill forms, submit information, automate repetitive web workflows
- **File & Document Management**: Create/edit documents, spreadsheets, presentations, organize file systems, convert file formats, generate reports
- **Data Processing & Analysis**: Clean/transform datasets, statistical analysis, create visualizations, monitor KPIs, generate insights, integrate data from multiple APIs/databases
- **System Administration**: Execute command-line operations safely, manage system configurations, automate DevOps workflows, monitor system health

Agent Modes

- **Basic Mode**: Core NYRA experience with basic autonomy
- **Advanced Mode**: Strongest autonomy & decision-making capabilities

Agent Types & Specializations

- **Email Assistant**: Smart inbox sorting, priority flagging, automated replies, email categorization, spam filtering
- **Daily Recap Worker**: Daily summaries, calendar overview, priority tasks, weather & news, team updates
- **Weekly Recap Worker**: Weekly achievements, goal tracking, upcoming priorities, team performance, strategic insights
- **Lead Generation Worker**: Prospect identification, lead qualification, contact enrichment, CRM integration, pipeline management
- **Meeting Research Worker**: Background research, company analysis, social media insights, meeting prep reports, conversation starters
- **Presentation Worker**: Slide creation, content structuring, visual design, template customization, export formats
- **Customer Service Agents**: Handle support tickets, FAQ responses, user onboarding, escalate complex issues, track satisfaction
- **Content Creation Agents**: Marketing copy, social media posts, technical documentation, educational content, content calendars

- **Sales & Marketing Agents**: Lead qualification, CRM management, meeting scheduling, personalized outreach, sales reports
- **Research & Development Agents**: Academic/scientific research, industry trend monitoring, patent analysis, research reports
- **Industry-Specific Agents**: Healthcare (patient data, appointments), Finance (risk assessment, compliance), Legal (document review, case research), Education (curriculum, assessment)

🛠️ CORE TOOLS & CAPABILITIES

File Management (`sb_files_tool`)

- Create, read, modify, delete files
- Organize files into directories/folders
- Convert between file formats
- Search through file contents
- Batch processing multiple files
- AI-powered intelligent file editing with natural language instructions
- HTML file preview with auto-exposed port 8080
- File permissions management
- Workspace state management

Browser Automation (`browser_tool`)

- Navigate to any URL
- Click buttons, links, interactive elements
- Fill out forms (text, numbers, emails)
- Select options from dropdowns and menus
- Scroll pages (up, down, to specific elements)
- Handle dynamic content and JavaScript-heavy sites
- Extract structured data from pages
- Take screenshots at any point
- Press keyboard keys (Enter, Escape, Tab, etc.)
- Handle iframes and embedded content
- Upload files (with filePath parameter)
- Navigate browser history (go back, forward)
- Wait for content to load
- Automatic screenshot validation for data entry
- Sandboxed environment for security

Web Search (`web_search_tool`)

- Internet search capabilities
- Information gathering
- Research topics
- Multi-source data collection

Shell/Command Execution (`sb_shell_tool`)

- Execute commands and run scripts
- System operations
- Development tasks
- Safe command-line execution

Vision & Image Processing (`sb_vision_tool`)

- Process images
- Analyze screenshots
- Extract text from images (OCR)
- Image analysis and understanding

Spreadsheet Operations (`sb_spreadsheet_tool`)

- Create interactive Excel (.xlsx) files
- Native Excel format with full formula support
- Cell formatting (colors, fonts, alignment)
- Auto-styled headers (dark blue with white text)
- Interactive viewing in chat interface
- Downloadable Excel files
- Multi-sheet workbooks
- Formulas (=SUM, =AVERAGE, =IF, etc.)

Presentation Creation (`sb_presentation_tool`)

- Generate professional HTML presentations
- Beautiful slide designs
- Multiple presentation templates (18+ styles)
- Custom theme creation
- Slide management (create, list, delete)
- Export presentations (HTML, PDF)
- Template customization
- Content structuring

- Visual design capabilities

Knowledge Base (`sb_kb_tool`)

- Semantic search on files using natural language queries
- Automatic file indexing in background
- Local knowledge base (sandbox files)
- Global knowledge base management
- CRUD operations for global KB (create folders, upload files, list contents, delete items, enable/disable)
- Sync global KB files to sandbox
- Check indexing status
- Cleanup operations (default, remove_files, clear_embeddings, clear_all)

Git Integration (`sb_git_sync`)

- Sync files and projects with Git repositories
- Version control
- Collaboration support

Task Management (`task_list_tool`)

- Create, view, update, delete tasks
- Organize tasks into sections
- Task status management
- Clear all tasks
- Batch operations

Canvas AI (`sb_canvas_tool`)

- AI-powered canvas/design capabilities

Designer Tool (`sb_designer_tool`)

- Design and visual creation capabilities

Document Parser (`sb_document_parser`)

- Parse and extract information from documents

File Reader (`sb_file_reader`)

- Read and process various file types

Image Edit (`sb_image_edit_tool`)

- Edit and manipulate images

Upload File (`sb_upload_file_tool`)

- Upload files to workspace

Expose Tool (`sb_expose_tool`)

- Expose local services
- Create public URLs for testing

Voice Integration (`vapi_voice_tool`)

- Voice call capabilities
- End calls
- Get call details
- Wait for call completion
- List calls

Search Tools

- **Company Search** (`company_search_tool`): Search for company information
- **People Search** (`people_search_tool`): Search for people information
- **Paper Search** (`paper_search_tool`): Academic paper search
- **Image Search** (`image_search_tool`): Image search capabilities

Specialized Tools

- **Apify Tool** (`apify_tool`): Web scraping and automation via Apify
- **Reality Defender Tool** (`reality_defender_tool`): Content verification
- **Message Tool** (`message_tool`): Messaging capabilities
- **Expand Message Tool** (`expand_msg_tool`): Message expansion

🔌 INTEGRATIONS & MCP SERVERS

Integration Platform

- **2700+ Integrations** via Composio MCP servers
- Search and discover integrations
- Credential profile management
- Secure authentication
- Tool selection and configuration
- Popular integrations discovery

Available Integration Categories

- **Communication**: Gmail, Outlook, Slack, Teams, Discord, SMS services
- **Project Management**: Notion, Asana, Linear, Todoist, Jira
- **Development**: GitHub, GitLab, CI/CD platforms
- **Storage**: Google Drive, Dropbox, cloud storage
- **Analytics**: Google Analytics, business intelligence platforms
- **CRM**: Salesforce, HubSpot, Apollo, ZoomInfo
- **Social Media**: Twitter, LinkedIn, Instagram
- **Databases**: Various database integrations
- **Design**: Canva, Unsplash, design tools
- **Presentations**: PowerPoint, Google Slides
- **Calendar**: Google Calendar, scheduling tools
- **News & Data**: News APIs, weather APIs, data providers

Integration Features

- Search MCP servers by keyword
- Get popular MCP servers
- Get MCP server tools and capabilities
- Test MCP server connections
- Create credential profiles
- Configure profiles for agents
- Discover user MCP servers
- Enable/disable specific tools per integration
- Secure credential storage

⏰ TRIGGERS & AUTOMATION

Scheduled Triggers

- Cron-based scheduling
- Daily, weekly, monthly schedules
- Custom time-based triggers
- Enable/disable triggers
- View all scheduled triggers
- Delete scheduled triggers
- Toggle trigger status

Event-Based Triggers (Composio)

- App-based event triggers
- List apps with available event triggers
- List triggers for specific apps
- Create event triggers
- Auto-respond to events (new emails, Slack messages, form submissions, etc.)
- Real-time event processing

Trigger Capabilities

- Run at specific times (9am daily, every Monday, first of month, etc.)
- Respond to external service events
- Multi-step automation workflows
- Conditional execution

🧠 MEMORY & CONTEXT

AI Memory System

- Automatic memory extraction from conversations
- Memory types: facts, preferences, context, conversation summaries
- Confidence scoring for memories
- Memory retrieval for context
- Memory management (view, search, delete)
- Memory limits per subscription tier
- Enable/disable memory feature
- Memory usage tracking

Memory Features

- Personal facts storage (name, role, company, location, interests)
- Preference tracking (communication style, tools, frameworks, languages)
- Project context (tech stack, goals, what they're building)
- Conversation summaries (key insights, decisions)
- Semantic search across memories
- Filter memories by type
- Clear all memories

📚 KNOWLEDGE BASE

Global Knowledge Base

- Upload files to knowledge base
- Organize files in folders
- Enable/disable files per agent
- Sync knowledge base to sandbox
- Semantic search across knowledge base
- File management (create, read, update, delete)
- Folder structure (1-level deep)

Local Knowledge Base

- Automatic indexing of workspace files
- Semantic search on indexed files
- Check indexing status
- Cleanup operations
- Path-specific search

🎨 TEMPLATES & PRESENTATIONS

Presentation Templates (18+ Styles)

- Architect
- Black and White Clean
- Colorful
- Competitor Analysis Blue
- Elevator Pitch
- Gamer Gray
- Green
- Hipster
- Minimalist
- Minimalist 2
- Numbers Clean
- Numbers Colorful
- Portfolio
- Premium Black
- Premium Green

- Professor Gray
- Startup
- Textbook

Template Features

- List available templates
- Load template designs
- Custom theme creation
- Slide management
- Export formats (HTML, PDF)
- Template metadata
- Preview capabilities

💳 BILLING & SUBSCRIPTION

Subscription Tiers

- **Basic (Free)**: 100 daily credits, 1 concurrent run, 10 total chats, Basic Mode
- **Plus ($20/month)**: 4,000 credits/month, unlimited chats, 3 concurrent runs, 5 custom AI Workers, 5 scheduled triggers, 25 app triggers, 100+ integrations, Advanced mode
- **Pro ($50/month)**: 10,000 credits/month, unlimited chats, 5 concurrent runs, Advanced mode
- **Ultra ($200/month)**: 40,000 credits/month, unlimited chats, 20 concurrent runs, Advanced mode

Billing Features

- Monthly and yearly billing options
- 15% yearly discount
- 7-day free trial
- Credit system (daily refresh, monthly allocation, extra credits)
- Credit purchase
- Subscription management
- Cancel subscription
- Upgrade/downgrade plans
- Trial management
- Billing history
- Usage tracking
- Credit balance monitoring
- Low credit warnings

Payment Providers

- Stripe integration
- RevenueCat (for mobile apps)
- Local mode (billing disabled for development)

👥 USER MANAGEMENT & AUTHENTICATION

Authentication

- Email/password authentication
- GitHub OAuth
- Phone verification
- Password reset
- Account activation
- Session management
- JWT tokens

User Features

- Profile management
- Avatar upload
- Account settings
- API key management
- Credential management
- Team management (if applicable)

📊 ADMIN FEATURES

Admin Dashboard

- User management
- Billing administration
- Analytics dashboard
- Feedback management
- Notification management

Analytics

- Daily/weekly signups

- Subscription conversion rates
- Thread analytics with message count filters
- AI-powered thread summarization
- Retention metrics
- User activity tracking
- Revenue analytics
- ARR simulator
- Thread browser with filters
- User & billing management
- Google Analytics integration
- Vercel analytics integration

Admin Capabilities

- View all users
- Manage user credits
- View billing details
- Access user feedback
- Send notifications
- View system metrics
- Debug information
- Queue metrics
- Worker metrics

🔔 NOTIFICATIONS

Notification System

- In-app notifications
- Email notifications
- Push notifications (mobile)
- Notification preferences
- Notification history
- Admin notification management

Presence System

- User presence tracking
- Online/offline status

- Real-time presence updates

📱 MOBILE APP FEATURES

Mobile Platforms

- iOS app
- Android app
- React Native/Expo

Mobile Features

- Chat interface
- Agent management
- Thread/conversation management
- Worker configuration
- Trigger management
- Billing & subscription
- Usage tracking
- Settings
- Onboarding flow
- Authentication
- Drawer navigation
- Agent selection
- Thread loading
- Profile management

💻 DESKTOP APP FEATURES

Desktop Application

- Electron-based desktop app
- Cross-platform support (macOS, Windows, Linux)
- App icons and branding
- Code signing (macOS)
- Notarization (macOS)

🌐 FRONTEND FEATURES

Web Dashboard

- Next.js/React application
- Responsive design
- Dark mode support
- Real-time updates
- Chat interface
- Agent configuration dashboards
- Workflow builders
- Monitoring tools
- Deployment controls

Pages & Routes

- Home page
- Dashboard
- Agents page
- Agent configuration (instructions, tools, integrations, knowledge, triggers, workflows)
- Thread/conversation pages
- Projects page
- Knowledge base page
- Triggers page
- Settings (API keys, credentials)
- Billing/subscription page
- Admin pages (analytics, billing, feedback, notifications)
- Help & support
- About page
- Careers page
- Enterprise page
- Status page
- Legal page
- Share pages (public thread sharing)
- Template pages
- Onboarding demo
- Credits explained page

UI Components

- Chat components
- Agent cards
- Thread list
- File attachments
- Tool views
- Feedback drawer
- Loading states
- Error handling
- Toast notifications
- Modals and dialogs
- Data tables
- Charts and graphs
- Calendar components
- Form components

🔧 AGENT CONFIGURATION

Agent Builder

- Create new agents
- Configure agent name, description, system prompts
- Set agent icons (100+ options)
- Configure tool access
- Add/remove integrations
- Set up knowledge base
- Configure triggers
- Define workflows
- Custom personality and expertise
- Agent cloning
- Agent deletion

Configuration Screens

- Instructions screen (system prompt editing)
- Tools screen (enable/disable tools)
- Integrations screen (MCP server management)
- Knowledge screen (knowledge base management)

- Triggers screen (scheduled and event triggers)
- Workflows screen (workflow configuration)

🔐 SECURITY & CREDENTIALS

Credential Management

- Secure credential storage
- Credential profiles
- OAuth integration
- API key management
- Connection link generation
- Account connection flow
- Profile management
- Secure MCP connections

Security Features

- Sandboxed execution environment
- Rate limiting
- Authentication middleware
- CORS protection
- Input validation
- Secure file handling
- Environment variable management

📈 MONITORING & METRICS

System Monitoring

- Health checks
- Docker health checks
- System metrics endpoint
- Queue metrics (Redis Streams)
- Worker metrics
- Memory watchdog
- Debug information
- Instance ID tracking

Metrics

- Queue pending messages
- Worker count and utilization
- Memory usage
- Request tracking
- Error logging
- Performance metrics

🧪 TESTING & DEVELOPMENT

Test Harness

- Agent evaluation
- Test case management
- External datasets
- Scoring system
- Simple test runner
- Agent eval runner

Development Features

- Local development mode
- Hot reload
- Debug mode
- Build verification
- Script utilities
- Development tools

🌍 INTERNATIONALIZATION

Supported Languages

- English (en)
- German (de)
- Spanish (es)
- French (fr)
- Italian (it)
- Japanese (ja)

- Portuguese (pt)
- Chinese (zh)

i18n Features

- Translation files
- Language switching
- Localized content
- Date/time formatting
- Number formatting

🔄 VERSIONING & DEPLOYMENT

Versioning

- API versioning
- Agent versioning
- Configuration versioning
- Migration support

Deployment

- Docker support
- Docker Compose
- Kubernetes ready
- Environment configuration
- Setup wizard
- Automated deployment

📝 DOCUMENTATION & HELP

Documentation

- README files
- API documentation
- Setup guides
- Contributing guidelines
- Feature documentation
- Tool usage guides

Help Features

- Help page
- Credits explained
- Support page
- FAQ (if applicable)
- Status page

🎁 REFERRALS

Referral System

- Referral program
- Referral tracking
- Referral rewards

📤 SHARING & COLLABORATION

Sharing Features

- Public thread sharing
- Share page generation
- OG image generation
- Share links
- Template sharing

🔍 SEARCH & DISCOVERY

Search Capabilities

- Semantic search (knowledge base)
- File content search
- Memory search
- Integration search
- Agent search
- Thread search

🎯 CATEGORIZATION

Categorization System

- Thread categorization
- Automatic categorization
- Category management

📞 SUPPORT & FEEDBACK

Support Features

- User feedback collection
- Feedback management (admin)
- Support page
- Contact forms
- Issue reporting

🚀 PERFORMANCE & SCALABILITY

Performance Features

- Redis Streams for task queuing
- Worker concurrency
- Background task processing
- Caching (runtime cache)
- Database connection pooling
- Async operations
- Rate limiting
- Memory optimization

Scalability

- Horizontal scaling support
- Load balancing ready
- Auto-scaling metrics
- Queue-based architecture
- Distributed locking

🎨 BRANDING & CUSTOMIZATION

Branding

- NYRA brand colors (black/white)
- Custom icons
- Logo management
- Brandmark effects
- Custom themes

UI Customization

- Dark mode
- Light mode
- Custom color schemes
- Responsive layouts
- Custom fonts

🔗 API & INTEGRATIONS

REST API

- FastAPI backend
- OpenAPI/Swagger documentation
- API key authentication
- Bearer token authentication
- Rate limiting
- Request logging
- Error handling

Webhooks

- Webhook support
- Trigger webhooks
- Integration webhooks

External APIs

- Google Slides API
- Google Docs API
- Transcription API

- Canvas AI API
- Various MCP server APIs

📦 FILE HANDLING

File Operations

- File upload
- File download
- File staging
- File preview
- File conversion
- File organization
- File search

🎓 ONBOARDING

Onboarding Features

- Setup wizard (14 steps with progress saving)
- Onboarding demo
- Guided tour
- First-time user experience
- Agent selection guide
- Integration setup guide

🏢 ENTERPRISE FEATURES

Enterprise

- Enterprise page
- Custom enterprise solutions
- Team management
- Advanced security
- Dedicated support

📊 USAGE TRACKING

Usage Features

- Credit usage tracking
- Daily credit refresh
- Monthly credit allocation
- Usage limits
- Usage history
- Usage analytics
- Concurrent run limits

🎪 SPECIAL FEATURES

Special Capabilities

- Thread manager
- Sandbox execution environment
- Docker containerization
- Browser automation in sandbox
- Code interpreter
- File system access
- Tool integration
- Security sandboxing
- Scalable agent deployment

🔄 WORKFLOWS

Workflow Features

- Workflow builder
- Multi-step automation
- Conditional logic
- Workflow templates
- Workflow execution
- Workflow monitoring

📋 PROJECTS & ORGANIZATION

Project Management

- Project creation
- Project organization
- Thread organization by project
- Project-based access control

🎯 ENDPOINTS

Custom Endpoints

- Endpoint creation
- Endpoint management
- Custom API endpoints

🧩 COMPOSIO INTEGRATION

Composio Features

- 2700+ tool integrations
- Toolkit service
- Integration service
- MCP server service
- Event triggers
- Credential management
- Tool discovery

📡 REAL-TIME FEATURES

Real-Time Capabilities

- Real-time chat
- Real-time updates
- WebSocket support (if applicable)
- Live agent monitoring
- Real-time presence

- Real-time notifications

🎨 DESIGN SYSTEM

Design Components

- UI component library
- Consistent styling
- Responsive design
- Accessibility features
- Animation support
- Loading states
- Error states
- Empty states

🔒 PRIVACY & COMPLIANCE

Privacy Features

- Data encryption
- Secure storage
- Privacy controls
- Data deletion
- GDPR compliance (if applicable)
- User data management

🎯 CATEGORIZATION & ORGANIZATION

Organization Features

- Thread categorization
- Agent categorization
- File organization
- Knowledge base organization
- Tag system
- Filtering and sorting

🚨 ALERTS & WARNINGS

Alert System

- Low credit warnings
- Subscription expiration alerts
- Error notifications
- System alerts
- User notifications

📱 RESPONSIVE DESIGN

Responsive Features

- Mobile-responsive web app
- Tablet optimization
- Desktop optimization
- Touch-friendly interfaces
- Adaptive layouts

🎁 BONUS FEATURES

Additional Features

- Daily credit refresh (100 credits for all tiers)
- Concurrent run management
- Chat limits (free tier: 10 chats)
- Unlimited chats (paid tiers)
- Custom AI Workers (paid tiers)
- Scheduled triggers (paid tiers)
- App triggers (paid tiers)
- Integration access (paid tiers)
- Advanced mode (paid tiers)

🔄 UPDATES & MAINTENANCE

Maintenance Features

- Graceful shutdown
- Health checks
- System monitoring
- Error recovery
- Logging and debugging
- Performance optimization
- Memory management

Total Feature Count: 500+ individual features across all categories

This comprehensive list covers every feature, capability, and functionality available in the NYRA platform. The platform is designed as a complete AI agent development and deployment system with extensive customization, integration, and automation capabilities.